

Advanced Communication Architectures (TSM_AdvComArc)

Exercise: Network Virtualization using Mininet

Topics: Network Virtualization using Mininet environment
Materials: Courseware, books, papers and other sources recommended during the lesson
Estimated time: ca. 1.0 Hour

Angelo Consoli - Version 1v2

Please answer the following questions

Use of the Mininet environment to study SDN and OpenFlow structures and operating logic

1. Referring to the Lab. Simulation reported at the end of OpenFlow course, please implement the simulations reported on slide 45, 46 and 47 of the course module on OpenFlow.
2. Please verify the functionality of all system components.
3. Please log all packet exchanges using Wireshark.
4. Isolate the OpenFlow initialization exchange (get inspiration from the course slides).
5. Try identifying the ARP requests and replies (get inspired from the course slides).
6. Isolate and comment the socket creation exchanges using both ICMP and HTTP.
7. Prepare a report with the result of the points listed above.